Technical Data Sheet

(TDS)

TINMORRY PPA-CF

Version: V1.0

• Introduction

TINMORRY PPA-CF is a high-performance, carbon-fiber-reinforced, high-temperature nylon. It offers excellent strength, stiffness, heat resistance, oil resistance, and durability, together with notably high toughness and interlayer bonding strength.

Note: 1. Because PPA-CF contains carbon fiber, use a nozzle diameter ≥ 0.4 mm to reduce the risk of nozzle clogging. In addition, use a wear-resistant nozzle such as hardened steel. Do not use softer nozzles such as brass or stainless steel, as they may wear rapidly. 2. An enclosed printer is strongly recommended to reduce warping and ensure sufficient interlayer strength.

• Product Specifications

Item	Data
Filament Diameter	1.75 ± 0.03 mm
Net Weight	1000 g ± 0.5%
Length	350 ± 5 m
Spool Material	PP material (heat resistant to approx. 100 °C)
Spool Dimensions	Outer diameter: 200 mm; hub opening diameter: 55 mm; Width: 62 mm

• Recommended Printing Parameters

Pre-print Drying Conditions	100-140 °C, 8-12 h (forced-air oven or filament dryer)
Printing Environment	≤ 75 °C, ≤ 20% RH (during printing, keep sealed in a container with desiccant, or use a dry box set to approx. 70 °C)
Storage Conditions	≤ 30 °C, ≤ 20% RH (keep sealed with desiccant for moisture protection)
Compatible Support Material	Same material or dedicated PA support material
Compatible Printer Type	Enclosed (recommended)
Compatible Nozzle Material	Hardened steel or other wear-resistant materials
Compatible Nozzle Diameter	0.4, 0.6 (recommended), 0.8 mm
Compatible Build Plate	Textured PEI build plate, etc.
Build Plate Surface Preparation	Applying glue or adhesive spray is recommended
Recommended Nozzle Temperature	280-300 °C
Heated Bed Temperature	90-110 °C (depending on printer and build plate type)
Printing Speed*	Recommended: ≤ 100 mm/s
Chamber Temperature	< 80 °C

*When printing with a Bambu Lab X1C or P1S at a nozzle temperature of 290 °C, the maximum printing speed can reach 300 mm/s.

• Properties

The commonly used physical, mechanical, chemical, and printing properties of TINMORRY PPA-CF have been tested and are listed in the tables below. Some specimens were annealed before testing using the following process: 100 °C for 4 h → 120 °C for 4 h.

• Physical Properties

Item	Test Method	Data
Density	ρ = m / v	1.25 g/cm³
Saturated Moisture Absorption	25 °C, 55% RH, ambient air	1.3%
Melt Flow Index	280 °C, 2.16 kg	8.4 ± 0.7 g/10 min
Glass Transition Temperature	DSC, 10 °C/min	85 °C
Crystallization Temperature	DSC, 10 °C/min	175 °C
Melting Temperature	DSC, 10 °C/min	252 °C
Vicat Softening Temperature	ISO 306, GB/T 1633	227 °C

Heat Deflection Temperature 1	ISO 75, 1.8 MPa	190 °C
Heat Deflection Temperature 2	ISO 75, 0.45 MPa	220 °C

• Mechanical Properties

Item	Test Method	Data
Tensile Strength XY	ISO 527, GB/T 1040	175 ± 4 MPa
Tensile Strength Z	ISO 527, GB/T 1040	48 ± 45 MPa
Young's Modulus XY	ISO 527, GB/T 1040	11530 ± 520 MPa
Young's Modulus Z	ISO 527, GB/T 1040	4380 ± 230 MPa
Elongation at Break XY	ISO 527, GB/T 1040	3.5% ± 1%
Elongation at Break Z	ISO 527, GB/T 1040	1.2% ± 0.3%
Flexural Strength XY	ISO 178, GB/T 9341	212 ± 5 MPa
Flexural Strength Z	ISO 178, GB/T 9341	65 ± 3 MPa
Flexural Modulus XY	ISO 178, GB/T 9341	9753 ± 375 MPa
Flexural Modulus Z	ISO 178, GB/T 9341	3283 ± 230 MPa
Impact Strength XY	ISO 179, GB/T 1043	40.2 ± 2.6 kJ/m²
Impact Strength Z	ISO 179, GB/T 1043	5.2 ± 0.6 kJ/m²

• Other Physical and Chemical Properties

Item	Data
Main Composition	Nylon, carbon fiber
Skin Irritation	May cause allergic reactions in some individuals; gloves are recommended
Water Resistance	Resistant
Oil and Grease Resistance	Resistant to most common oils and greases
Organic Solvent Resistance	Not resistant to certain organic solvents
Weak Acid Resistance	Resistant
Strong Acid Resistance	Not resistant
Weak Alkali Resistance	Resistant
Strong Alkali Resistance	Not resistant
Flammability	Flammable
Odor During Printing	Irritating odor
Odor During Combustion	Irritating odor

• Test Specimen Information

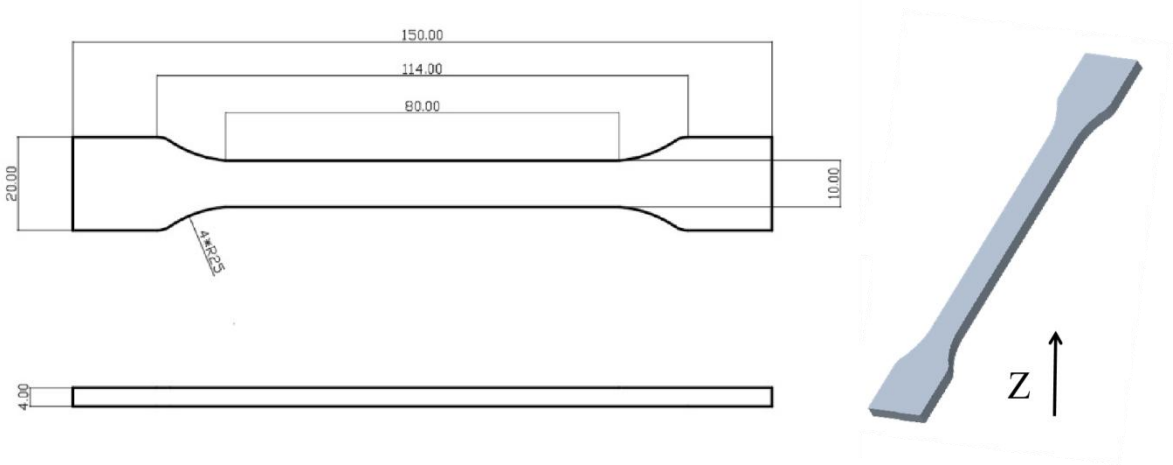
Mechanical Test Specimen Information	
Item	Data
Specimen Dimensions	See figure below
Nozzle Temperature	290 °C
Build Plate Temperature	100 °C

Printing Speed	XY: 100 mm/s; Z: 80 mm/s
Infill Density	100%
Infill Pattern	Concentric

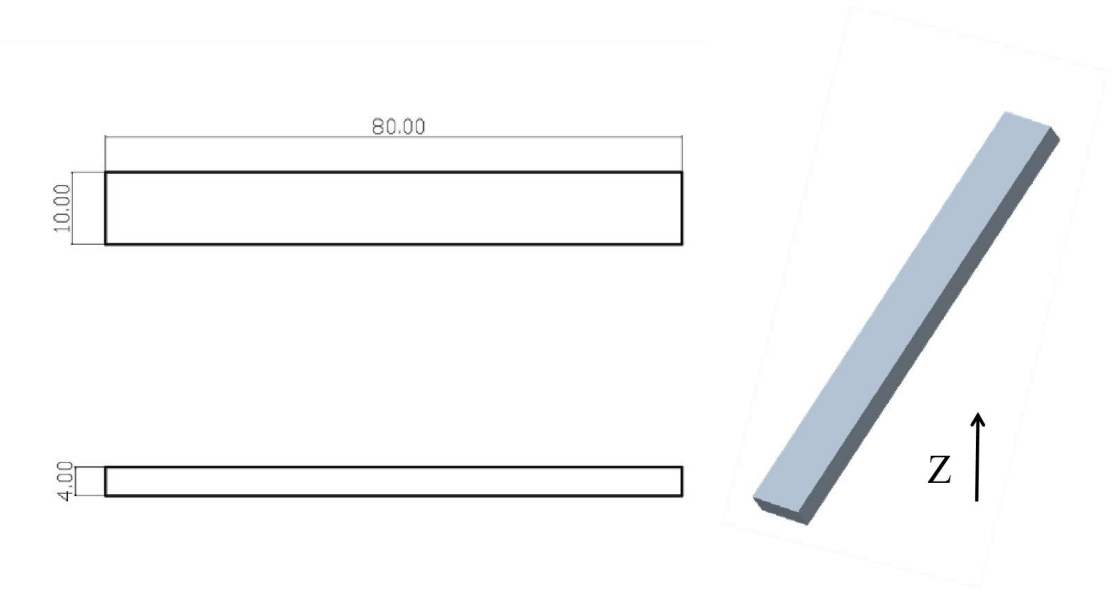
Notes:

1. The test specimens were printed using the key parameters listed above. Before testing, all specimens were conditioned indoors at approximately 25 °C for more than 12 h. Different ambient temperatures, fan speeds, and layer times during printing produce different cooling rates, which can affect the mechanical properties of the specimens, especially those printed in the Z direction. Different printers may also result in differences in the quality and performance of printed parts.
2. For applications requiring higher print quality, thoroughly dry TINMORRY filament before printing and protect it from moisture during printing using a sealed container with desiccant (recommended relative humidity inside the container ≤ 20%) to reduce moisture-related printing issues. Refer to the "Recommended Printing Parameters" table for the recommended drying conditions. During drying, keep the filament and spool away from direct heat sources to avoid overheating damage. Do not use a kitchen oven or microwave oven.
3. Printed parts made from TINMORRY PPA-CF may show a slight improvement in overall mechanical performance after annealing, but some parts may shrink or deform during the process. The recommended annealing procedure is to anneal at 100 °C for 4 h, then slowly raise the temperature to approximately 120 °C (not recommended to exceed 140 °C) and anneal for about 6 h, followed by slow cooling in the oven. Annealing behavior varies with part size, structure, and geometry. If obvious deformation occurs during the process, use a lower annealing temperature.

1. Tensile Test



2. Flexural Test and Unnotched Impact Test



• Disclaimer

The data above were obtained by TINMORRY using specific printers, test methods, industry-standard specimens and methods, and specific equipment, and are provided for reference and comparison only. If the data change, TINMORRY will update this TDS, but may not be able to provide individual notice to users. In addition, the performance of 3D-printed parts depends not only on the intrinsic properties of the filament, but also on many other factors, including filament moisture content, printer type and condition, ambient temperature and ventilation, part geometry, and printing parameters. Users are responsible for the safety, legality, regulatory compliance, and performance of parts produced using TINMORRY 3D printing filament. TINMORRY is responsible for the quality of the filament sold, but assumes no responsibility for any damage or injury arising during its use or for the suitability of the filament for any particular application.